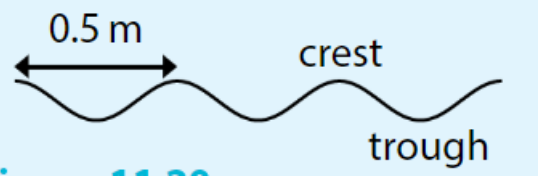


Relationship Between Period, Wave Speed, Frequency and Wavelength

Definition:	Explanation
Period, T is the time taken by each point on the wave to complete one oscillation SI unit is second (s) #PhysicsDefinitions	period is equivalent to the time taken for the wave to travel through a distance equal to one wavelength (you can also use the distance between 1st time it hits the mean and the 2nd time it returns bac to the mean position)
Frequency f is the number of oscillations each point completes per second SI unit is hertz (Hz) #PhysicsDefinitions Wavelength λ is the shortest distance between 2 successive crests or troughs SI unit is meter (m)	Frequency of a wave is also the number of crests (or troughs) that go past a point per second Frequency and period are also related by this equation $f = \frac{1}{T}$ #physicsFormulae $\Rightarrow$ T here is the period of the wave. 3 wave cycles are produced and move away from the source in 1s  Frequency $f = 3\text{Hz}$ Period $T = \frac{1}{3}\text{s}$
wave speed v is the distance travelled by the wave per second. SI unit is meter/second (m/s)	Recall that speed = $\frac{\text{distance}}{\text{time}}$. since every crest advances by 1 wavelength in one period, T the speed of the wave is given by $V = \frac{\lambda}{T}$ where v = wave speed (m/s) λ = wavelength (m) T = period (s) #physicsFormulae Since $f = \frac{1}{T}$, $v = f\lambda$ in the above figure $\Rightarrow$ wavelength of the wave is 0.5m Wave speed: $v = 0.5 \div \frac{1}{3} = 1.5\text{m/s}$ to conclude, relationship between v, f and λ is $v = f\lambda$ #physicsFormulae

Wave speed v is determined by *properties of the medium*

Wave speed on a rope is determined by factors like tension and thickness of rope.

Therefore for a wave that stays in the same medium, v is fixed.

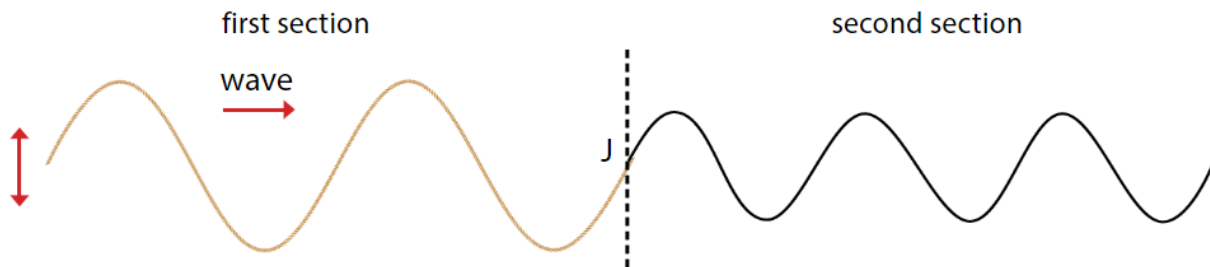
equation $v = f\lambda$ tells us that if the frequency f of the wave generator is increased, λ will decrease

sometimes waves can travel into a diff medium and its speed in a new medium can be higher or lower

however $\Rightarrow$ frequency of wave will always remain the same as a wave travels from one medium to another

if wave travels into a different medium such that v decreases, then λ will decrease since f remains the same

$\Rightarrow$ this is the figure to understand why frequency of wave remains the same as a wave travels from 1 medium to another.



At point J where 2 sections of ropes are joined $\Rightarrow$ oscillation of first section is passed on to the next section.

$\Rightarrow$ hence frequency f of oscillation in the second rope must be the same as in the first rope.

compare the wavelength in the second section to the first section and applying the formula $v = f\lambda$.
